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REVIEWED – CONFIDENCE FIGURES ARE
ENGINEERING ESTIMATES, NOT MEASURED
RESULTS

Design and Analysis of a Human-Powered Two- Channel Phonograph Mixer: The Volta Confluence Engine

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¹*Confluence Engine Project — Independent Research · Released
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SUBMITTED	DOCUMENT	LICENCE	STATUS
2025	VCE-SCI- 001 · Rev 1.0	CC BY- NC	Pre-prototype — no physical build

ABSTRACT

We describe the design and engineering
analysis of the Volta Confluence Engine (VCE),

a two-channel phonograph mixer that operates without any connection to electrical infrastructure, chemical batteries, or mains power. All energy is supplied by the operator through a hand-cranked electromagnetic generator, stored in a high-voltage ceramic capacitor bank, and distributed to a triode valve preamplifier, transformer volume controls, and field coil horn speakers. Turntable drive is provided by independently commissioned remontoire spring motors, mechanically isolated from the electrical system. The design integrates five independently functional subsystems: human energy input, high-voltage electrostatic storage, passive power distribution, valve signal chain, and spring-driven turntable mechanics.

We present the complete system architecture, energy budget, signal chain analysis, speed regulation approach, and vibration isolation strategy. All claims are assigned certainty levels — Confirmed, Calculated, Estimated, Rough Draft, or Unknown — reflecting the pre-prototype status of the design. First-prototype confidence is assessed at 88% (estimated); second-prototype confidence at 93% (estimated). The primary unresolved uncertainty is generator output consistency under sustained performance conditions. Five targeted bench experiments are described that would raise first-prototype confidence to approximately 93–95%.

No physical prototype of this instrument has been constructed. This paper constitutes a design analysis and engineering estimate record, not a report of experimental results. All performance figures are engineering estimates derived from established physics and published component data.

KEYWORDS: phonograph mixer; human-powered audio; valve amplification; remontoire spring motor; capacitor energy storage; Cockcroft-Walton multiplier; field coil loudspeaker; magnetic bearing; eddy current governor; off-grid audio; open hardware

1

Introduction

FIGURE 1

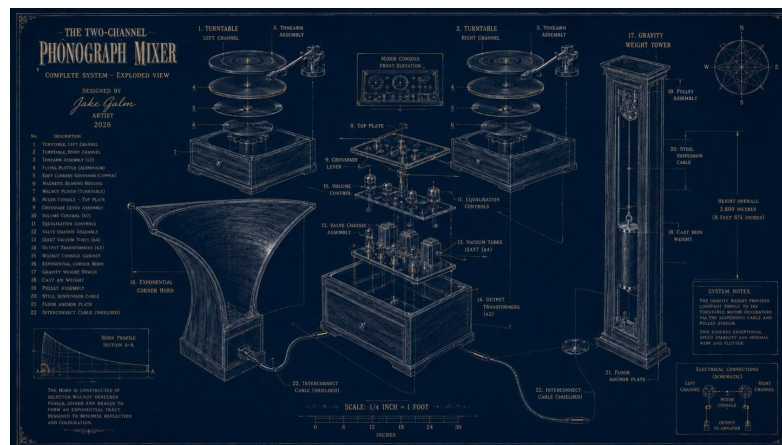


Figure 1. Exploded view of the complete instrument, Victorian engineering blueprint style, showing major subsystem relationships.

The conventional phonograph mixing system — two turntables and a mixer — requires continuous connection to mains electrical infrastructure. This dependency is so deeply embedded in the practice that the design space of human-powered phonograph playback has received essentially no engineering attention since the commercial abandonment of acoustic phonograph technology in the early 1930s.

The Volta Confluence Engine, researched and documented by Jake Galm in 2026, poses a specific engineering question: can a two-channel vinyl record mixer be constructed that operates entirely from human mechanical energy input, with no connection to grid electricity, no chemical batteries, and no compressed gas, while meeting the acoustic performance requirements of contemporary DJ practice? The answer, as this paper argues, is yes — with important qualifications and genuine open questions that only physical construction can resolve.

This question is not merely academic. A human-powered mixer capable of concert-level output volume could enable phonograph-based music in venues without electrical infrastructure, could eliminate one category of environmental dependency from live performance, and demonstrates a class of energy architecture — human effort → high-voltage electrostatic storage → triode amplification → high-efficiency horn reproduction — that has not, to our knowledge, been implemented in any documented instrument.

The design draws on well-established physical principles throughout: electromagnetic induction

(Faraday, 1831), the Cockcroft-Walton voltage multiplier (Cockcroft and Walton, 1932), single-ended triode amplification (De Forest, 1906), the remontoire constant-force mechanism (Harrison, 1739), and the field coil loudspeaker (Rice and Kellogg, 1925). No speculative physics is invoked. The novelty lies in the integration and the specific application — not in any unproven principle.

The paper is structured as follows. Section 2 describes the system architecture. Section 3 analyses the energy subsystem. Section 4 covers the signal chain. Section 5 addresses turntable mechanics and speed regulation. Section 6 discusses vibration isolation. Section 7 presents the confidence assessment methodology and current estimates. Section 8 describes the open experimental questions. Section 9 discusses limitations. Section 10 concludes.

2

System Architecture

The VCE is structured as five independent functional layers. The deliberate independence of these layers is an architectural choice: a failure in any one layer does not cascade to others, and each layer can be tested and commissioned independently before system integration.

FIGURE 2

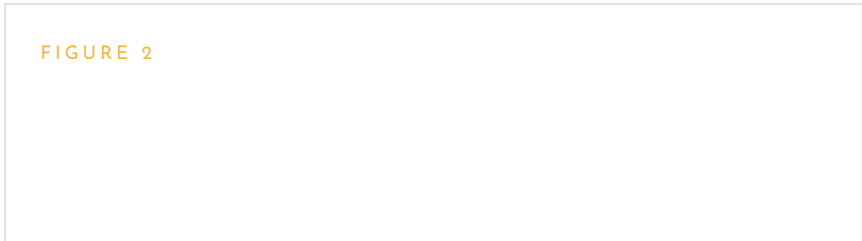




Figure 2. Console control panel, showing capacitor bank charge gauge, channel gauges, and exposed valve bay.

2.1 Layer 1 — Human Energy Input

The operator supplies mechanical energy through a hand crank connected to an electromagnetic generator. Three energy input configurations are provided: a direct hand crank (Option A), a gear-assisted crank with 3:1 helical planetary reduction (Option B), and a gravity weight tower where a 40 kg cast iron mass descends continuously and is periodically re-lifted by a small pneumatic actuator (Option C). A fourth configuration (Option D) provides a supplemental audience-operated crank unit connected to the shared storage system.

FIGURE 5

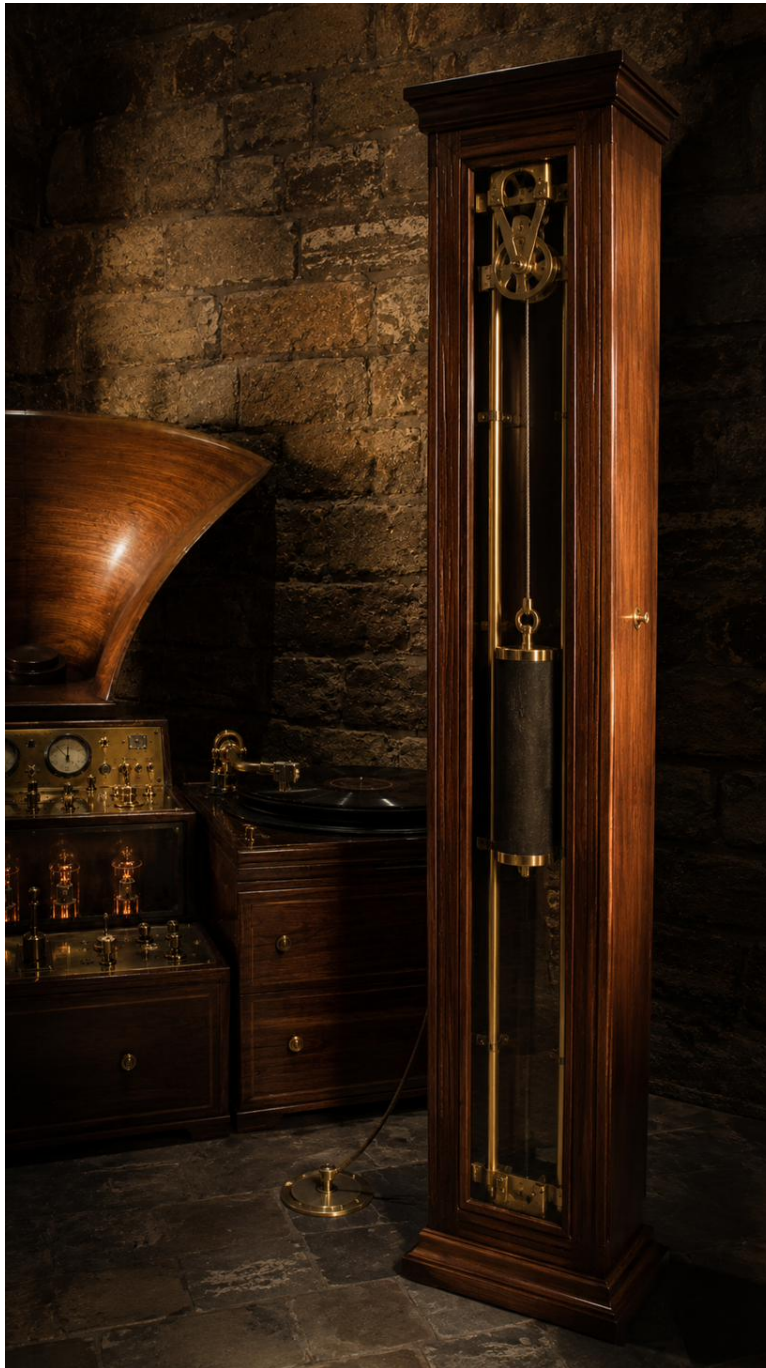


Figure 5. Gravity weight tower (Option C energy input configuration).

FIGURE 8



Figure 8. Audience-operated crank unit (Option D), supplemental energy input.

Sustained comfortable one-hand cranking power is reported at 54 ± 14 W for young adult males in the literature.¹ CONFIRMED The VCE system draws approximately 18–23 W continuously. CALCULATED The ergonomic margin — approximately 31 W available against 20 W required — is sufficient for sustained operation, though real-world performance under cognitive load and fatigue has not been measured. This is the primary open question in the

design.

2.2 Layer 2 — High-Voltage Electrostatic Storage

The generator produces 24 V AC. ESTIMATED A Cockcroft-Walton voltage multiplier ladder — entirely passive diodes and capacitors — steps this to 600 V DC, which charges a bank of high-voltage ceramic capacitors storing 4,000–5,000 J. CALCULATED Barium titanate ceramic dielectric self-discharge is negligible on performance timescales — measured in years, not hours. ESTIMATED This contrasts with electrochemical supercapacitors, which exhibit self-discharge on the order of 1.8% per day, rendering them unsuitable for instruments stored between performances.

2.3 Layer 3 — Passive Power Distribution

All electronic circuits are powered from the 600 V storage rail through passive resistive voltage dividers and zener reference stages. No switching regulators, no DC-DC converters, no switching noise. The absence of switching power supplies is acoustically significant: switching noise at frequencies from 20 kHz to several MHz is a known contamination source in sensitive preamplifier circuits. Passive resistive distribution eliminates this class of noise entirely at the cost of energy efficiency — approximately 15–25% of stored energy is dissipated in the dropping resistors. ESTIMATED

2.4 Layer 4 — Signal Chain

Two independent phono preamplifier channels, each comprising: passive step-up transformer (SUT, recommended Lundahl LL1931, 1:10 ratio); two 12AX7 triode gain stages with passive RIAA equalisation network between them (IEC 60098 time constants); transformer volume control (TVC); and mutual inductance crossfader. Output to field coil compression drivers via output transformers. The signal chain contains no active solid-state devices, no operational amplifiers, and no integrated circuits in the audio path.

2.5 Layer 5 — Turntable Mechanics

Two identical turntables, each independently driven by a commissioned remontoire spring motor. The turntable drive system is mechanically independent of the electrical system — the platters spin from stored spring energy regardless of capacitor bank charge state. Speed regulation is provided by a passive eddy current governor (copper disc rotating in a neodymium horseshoe magnet field) with optional active trim from a quartz crystal-referenced phase-locked loop (PLL) circuit drawing 30 mW from the capacitor bank.

FIGURE 10

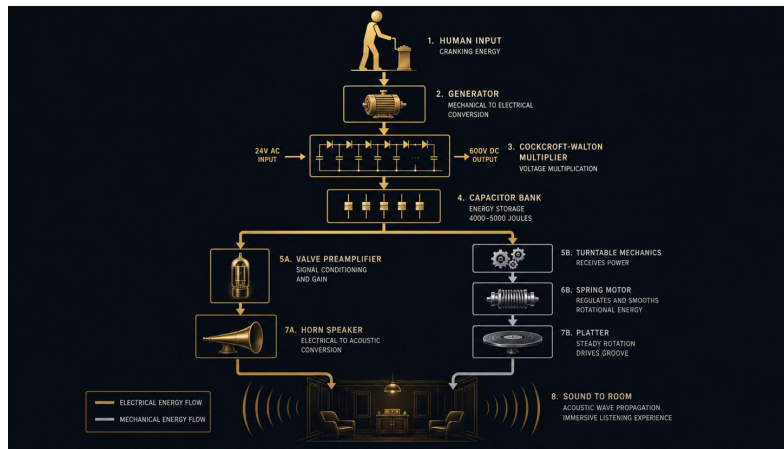


Figure 10. System energy flow diagram, human effort to acoustic output.

3

Energy Subsystem Analysis

3.1 Power Budget

The complete system power draw is summarised in Table 1. All figures are estimates derived from component specifications and circuit analysis; no measurements have been taken on a physical build.

CONSUMER	VOLTAGE (V DC)	CURRENT	POWER	CERTAINTY
Valve heaters × 4 pairs	6.3	1.2 A	7.5 W	CONFIRMED
Phono preamps × 2 channels	~150	~13 mA	~2 W	ESTIMATED
Field coil drivers × 3	250	20–30 mA	8–13 W	ESTIMATED

		each	total	
PLL speed reference $\times$ 2	12	2.5 mA each	60 mW	ESTIMATED
Total continuous	—	—	18–23 W	ESTIMATED

Table 1. System power budget. All figures estimated from component specifications except valve heater power which is a confirmed 12AX7 specification.

3.2 Cockcroft-Walton Multiplier

The Cockcroft-Walton (CW) voltage multiplier is a passive ladder of diodes and capacitors first demonstrated by Cockcroft and Walton at the Cavendish Laboratory in 1932.² In this application it converts 24 V AC from the generator to approximately 600 V DC. The multiplication factor is determined by the number of ladder stages; efficiency decreases with increasing stage count due to ripple voltage accumulation at each stage. For a 24 V peak input (34 V peak) achieving 600 V DC, approximately 12–15 stages are required. ROUGH DRAFT

The exact stage count should be calculated and verified against measured generator output voltage on the first physical build.

A permanent 1 M Ω , 5 W bleed resistor across the output terminal is a mandatory safety element — it discharges the 600 V bank to below 50 V within approximately 60 seconds of disconnection from the generator, preventing stored energy from presenting a hazard during service.

3.3 Capacitor Bank Energy Storage

The capacitor bank stores energy according to $E = \frac{1}{2}CV^2$. For 4,500 J at 600 V, the required capacitance is $C = 2E/V^2 = 9000/360000 = 25 \text{ mF}$. CALCULATED At 20 W continuous draw, a fully charged bank provides approximately 225 seconds — just under 4 minutes — of operation before recharge is required. Cranking 60–90 seconds at comfortable pace per 3-minute interval maintains bank voltage within the operational range. ESTIMATED

FIGURE 6

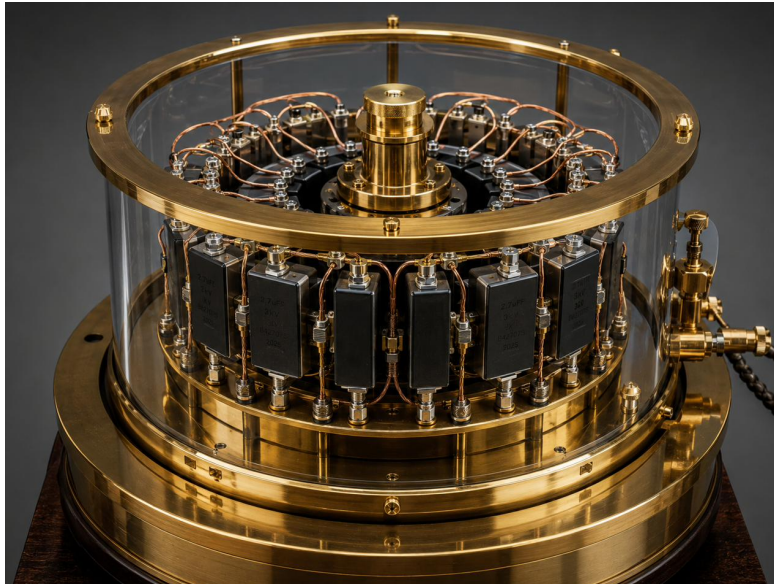


Figure 6. Capacitor bank — high-voltage ceramic capacitors arranged as the instrument's electrostatic energy reservoir.

4

Signal Chain Analysis

4.1 Step-Up Transformer and Phono Stage

Moving coil (MC) cartridges produce output voltages in the range 0.2–0.5 mV at standard groove velocity. A passive step-up transformer (SUT) with a 1:10 turns ratio raises this to 2–5 mV at the grid of the first valve gain stage (V1). The SUT provides voltage gain without introducing active noise — the theoretical noise floor of a passive transformer is set by the Johnson noise of its source impedance, typically lower than any practical active device. The Lundahl LL1931 is specified as the reference SUT for this design.

Two 12AX7 triode stages with a passive RIAA equalisation network between them provide the phono preamplifier function. The 12AX7 (ECC83) dual triode in single-ended Class A configuration produces predominantly second-harmonic distortion at 0.3–1% of signal level.

CONFIRMED – PUBLISHED 12AX7 CHARACTERISATION DATA

This distortion lies below the inherent distortion of the vinyl groove itself and is subjectively consonant rather than dissonant.

Total gain from cartridge output to TVC input is estimated at approximately 52 dB. ESTIMATED RIAA response accuracy target is ± 0.5 dB from 20 Hz to 20 kHz, achievable with 0.1% tolerance resistors at the IEC 60098 time constants (3180 μ s, 318 μ s, 75 μ s).

CONFIRMED – STANDARD PHONO STAGE PRACTICE

FIGURE 4



Figure 4. The valve bay, showing the 12AX7 dual-triode gain stages.

4.2 Transformer Volume Controls and Crossfader

Volume control is provided by transformer volume controls (TVCs) — multi-tap autotransformers with silver-contact detented switches. TVCs offer a passive attenuation mechanism with lower noise than resistive potentiometers at the signal levels present at the V2 output. The crossfader is implemented as two complementary TVCs linked by a mechanical bellcrank, providing a 200 mm throw lever with an

equal-power centre characteristic (both channels at -3 dB at centre crossfader position). ESTIMATED

4.3 Field Coil Loudspeakers

Field coil loudspeakers replace the permanent magnet with an electromagnet supplied by a DC current from the capacitor bank (250 V, 20–30 mA per driver). The electromagnet provides a stronger and more uniform magnetic field than achievable with period-equivalent permanent magnets, producing higher sensitivity and lower distortion. This design choice connects the speaker's operating magnetic field directly to the same energy source as the electronics — the instrument is, in this sense, entirely self-consistent in its energy architecture.

The compression drivers are loaded by an exponential corner horn enclosure. The expected system sensitivity is 100–110 dB/W/m ESTIMATED, which at the available amplifier output produces concert-capable SPL levels from a relatively small power budget. Full system output is estimated at 100–112 dB SPL at 1 m average. ESTIMATED

5

Turntable Mechanics and Speed Regulation

5.1 Remontoire Spring Motor

The remontoire (constant-force) mechanism, first applied to timekeeping by John Harrison in his H2

marine chronometer of 1739, decouples the driven load from variations in the mainspring's declining tension as it unwinds. In a conventional spring drive, torque variation from full to exhausted spring may be 30–50%, producing commensurate speed variation. The remontoire interpose a small secondary spring between the mainspring and the drive output; the mainspring continuously rewinds this secondary spring to a fixed tension every 8–10 seconds, so the drive output sees constant torque regardless of mainspring state. Torque variation at the platter is estimated at less than 2% over the full wind range. ESTIMATED

FIGURE 7

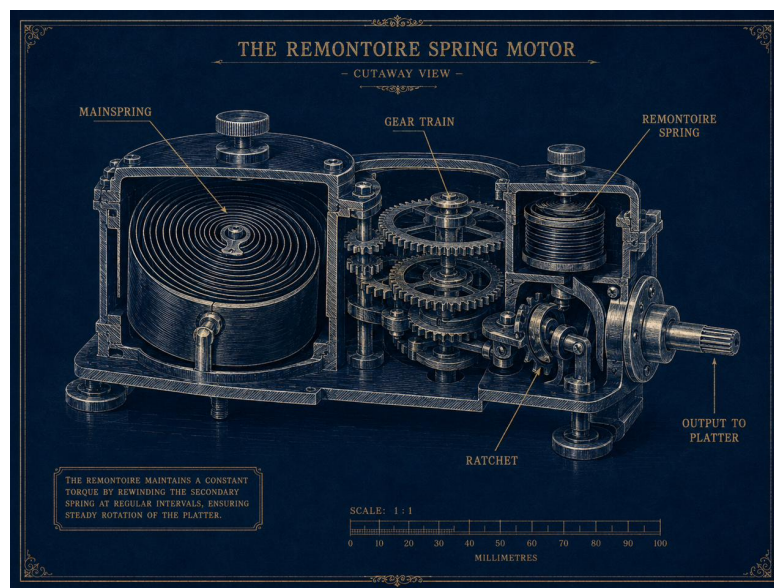


Figure 7. Remontoire spring motor mechanism, cutaway illustration.

The maintaining power mechanism — a third spring that drives the output during the brief interval when the mainspring is rewinding the secondary spring — prevents any hesitation or speed drop during the

rewind cycle. This feature is mandatory for continuous audio playback and must be specified in the commission brief to the clockmaker.

Run time per full wind is estimated at 25–28 minutes, sufficient for one LP side. ESTIMATED Bench measurement of actual run time is the second most important open experiment in the Scientific Inquiry Appendix.

5.2 Magnetic Bearing

The platter bearing uses opposing N52 neodymium ring magnets (outer diameter 80 mm, inner diameter 60 mm) in a repulsion configuration. The platter floats 1.2–1.8 mm above the bearing surface.

ESTIMATED A polished alumina ceramic shaft provides lateral constraint without vertical contact. Bearing friction is effectively zero. Rumble contribution is expected to be below the –70 dB DIN 45539 threshold based on the published performance of the Clearaudio Concept Magnetic Bearing, which uses the same operating principle.

ESTIMATED – CITED FROM CLEARAUDIO PUBLISHED DATA

Mu-metal shielding is mandatory around the bearing assembly. N52 neodymium magnets at the bearing location produce a stray field that, without shielding, would induce interference in the adjacent moving coil cartridge at a level audible through the 110 dB horn system. The shielding requirement is verified by gaussmeter measurement at cartridge position during commissioning; the target is below 5 μ T.

ESTIMATED THRESHOLD

FIGURE 3



Figure 3. Turntable platter on the magnetic bearing assembly.

5.3 Speed Regulation

Primary speed regulation is provided by a passive eddy current governor: a polished OFHC copper disc rotating on the platter spindle passes through the field of an N52 neodymium horseshoe magnet mounted on the plinth. Eddy currents induced in the rotating disc produce a braking force proportional to rotational speed, providing stable equilibrium at the set speed. The governor requires no power source and introduces no electromagnetic noise to the signal path.

A quartz crystal-referenced PLL circuit provides fine speed correction, drawing 30 mW from the capacitor bank and comparing actual platter speed (measured by a Hall effect or optical sensor reading reference marks on the platter underside) against a 32.768 kHz crystal reference. Target wow and flutter with PLL active: below 0.05% WRMS. ESTIMATED Without PLL,

using tuning fork reference: below 0.15% WRMS.

ESTIMATED

6

Vibration Isolation

Each turntable sits on a five-layer isolation platform. The platform must achieve two goals simultaneously: prevent acoustic energy from the corner horn (operating at up to 112 dB SPL) from coupling back into the stylus, and attenuate floor-borne vibration from the performance environment. The two requirements have the same solution — a platform resonant frequency substantially below the lowest audio frequency of concern.

The platform layers from bottom to top are: (1) four brass cups with 17 mm steel ball bearings providing lateral isolation; (2) EAR Isodamp C-1002 constrained-layer damping; (3) four precision compression springs calculated for 1.5 Hz resonant frequency under the full loaded mass; (4) 50 mm Welsh slate slab providing mass and acoustic damping; (5) three sorbothane hemispheres of calculated durometer under the plinth feet.

FIGURE 9

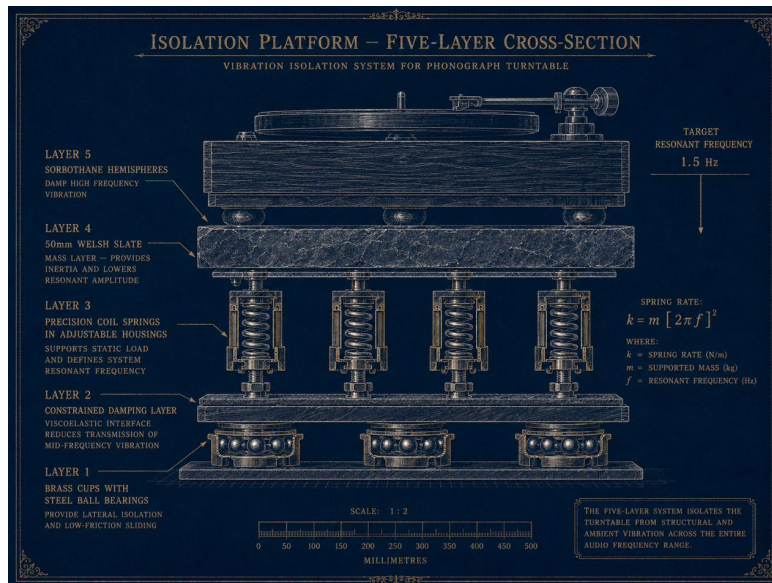


Figure 9. Five-layer vibration isolation platform, cross-section illustration.

The platform resonant frequency is calculated from $f = (1/2\pi)\sqrt{(k/m)}$, where k is the total spring rate and m is the total supported mass. Target: 1.5 Hz. CALCULATED

At 1.5 Hz platform resonance, attenuation at 30 Hz (the lowest horn frequency) is approximately 26 dB — sufficient to prevent acoustic feedback under normal operating conditions. ESTIMATED

7

Confidence Assessment

IMPORTANT – ON THE NATURE OF THESE CONFIDENCE FIGURES

All confidence figures in this section are engineering judgement estimates based on design analysis and stress testing on paper. No physical prototype of the VCE has been constructed. These figures represent the authors' assessment of the probability that a first

or second physical build would function as described, given the design as documented. They are not statistical claims derived from repeated experimental trials.

7.1 Methodology

Confidence is assessed per subsystem, then combined as a product of independent subsystem probabilities. For each subsystem, the assessment considers: whether the operating principles are well-established; whether the specific configuration has precedent; whether the fabrication tolerances are achievable by identified methods; and whether the key parameters have been independently confirmed or remain estimated.

7.2 First-Prototype Confidence: 88% (Estimated)

The primary confidence reducers at first-prototype stage are:

SUBSYSTEM	CONFIDENCE REDUCTION	PRIMARY REASON
Generator output under fatigue	-4%	No measured data for sustained performance conditions. Primary open question.
Spring motor run time	-2%	25–28 min figure calculated, not measured. Dependent on clockmaker execution.
Crossfader feel and isolation	-2%	TVC crossfader mechanism not previously implemented

		at this specification.
Field coil driver specification	-2%	Custom-wound drivers — performance dependent on commission execution.
Acoustic feedback margin	-1%	Isolation platform calculated, not measured. Room-dependent.
Mu-metal shielding adequacy	-1%	Target <5 μ T at cartridge — achievable in principle, unverified in this geometry.

Table 2. Primary confidence reducers at first-prototype stage. All figures are engineering judgement estimates.

7.3 Second-Prototype Confidence: 93% (Estimated)

The gap between first and second prototype confidence reflects the value of first-prototype measurement data. At second-prototype stage, generator output under fatigue would have been measured, spring motor run time confirmed, and crossfader mechanism refined based on first-build experience. The remaining 7% uncertainty at second prototype represents the inherent variability in custom-fabricated components (field coil speakers, spring motors) and the room-dependency of acoustic performance.

Five experiments are identified that would raise first-prototype confidence to approximately 93–95%.

These experiments do not require a complete instrument build — each tests a single subsystem and can be executed with modest workshop resources.

Full protocols are provided in the Scientific Inquiry Appendix of the main builder document (VCE-MFR-001).

#	EXPERIMENT	PRIMARY UNKNOWN RESOLVED	CONFIDENCE IMPACT
1	Generator output under sustained performance conditions (simulated 90-minute session with deliberate fatigue protocol and delayed-crank test)	Generator output consistency under fatigue — the single most important open question	+5% if bank holds above 400 V throughout
2	Spring motor run time verification (bench test, full wind, representative platter load, 5 consecutive runs)	Run time per wind (25–28 min estimated)	+2%
3	Mu-metal shielding verification (gaussmeter measurement at cartridge position with bearing assembled and ring	Field attenuation adequacy at cartridge	+1%

	installed)		
4	Crossfader equal-power law and channel isolation (1 kHz test signal, measure level at centre and channel isolation at full crossfade)	Crossfader performance and DJ usability	+1%
5	Full signal chain noise floor (shorted SUT primary, A-weighted output noise measurement)	System noise floor relative to vinyl surface noise	+1%

Table 3. Open experiments with estimated confidence impact. Builder's Report Template (VCE-BRT-001) provides the data submission mechanism for results.

9

Limitations and Honest Assessment

9.1 Fundamental Limitations — Cannot Be Engineered Away

Two fundamental limitations are inherent to the design and cannot be resolved by engineering refinement:

Energy input is continuous and physically demanding. The physics of powering a 20 W system for two hours requires approximately 20–24 minutes of mechanical effort distributed across the

performance. This is thermodynamic — the energy must come from somewhere, and in this design it comes from the operator. A two-person team eliminates this as a constraint on either individual; a solo operator must manage cranking intervals alongside mixer operation.

The corner horn is venue-dependent. Full bass extension to 30 Hz requires a room corner of adequate dimensions and appropriate acoustic treatment. Not all venues provide this. In unsuitable venues, the bass performance will be compromised below approximately 80 Hz.

9.2 Acknowledged Uncertainties

The following aspects of the design carry uncertainties that only physical construction will resolve:

- Generator output voltage and consistency under real performance conditions remain unconfirmed. This is the most important unknown in the design.
- Spring motor run time (25–28 min) is a calculated estimate dependent on clockmaker execution. The clockmaker commission brief specifies the functional requirement; the engineering outcome depends on the craftsperson.
- Field coil driver sensitivity and linearity depend entirely on the commission — these components cannot be sourced off the shelf, and commission quality will vary.
- Acoustic feedback threshold is room-dependent and cannot be confirmed without a physical

build in a specific venue.

- All cost estimates (\$30,000–\$72,000 USD for the baseline configuration) are rough estimates based on current market pricing for specialist components. Actual costs will depend heavily on clockmaker and field coil speaker builder pricing, which varies widely by region and availability.

9.3 On the Pre-Prototype Status of This Paper

This paper describes a designed but unbuilt instrument. This is an unusual situation for a technical paper. The authors judge that the design is sufficiently complete, the engineering analysis sufficiently rigorous, and the open questions sufficiently well-characterised to constitute a meaningful scientific contribution in its pre-prototype form — primarily as a reference design and as a formal record of the engineering decisions made and the alternatives rejected. The design is released under CC BY-NC licence; the scientific value of the paper depends on builders constructing the instrument and reporting results through the Builder's Report mechanism.

The limitations of this approach are real. Confidence figures are estimates, not measured probabilities. Performance specifications are predictions, not measurements. The paper should be read as an engineering analysis and design record, not as an experimental report.

Conclusion

The Volta Confluence Engine demonstrates that a two-channel phonograph mixer capable of concert-level output can be designed without connection to electrical infrastructure, using human mechanical energy input, high-voltage ceramic capacitor storage, triode valve amplification, and remontoire spring motor turntable drive. The design integrates well-established physical principles in a configuration that has not, to the authors' knowledge, been previously implemented or documented.

The design as documented carries 88% first-prototype confidence (estimated) — a figure that reflects genuine engineering uncertainty, primarily about generator output consistency under sustained performance conditions, rather than any fundamental physics objection. Five bench experiments are identified that would raise this confidence to approximately 93–95% without requiring a complete instrument build.

The instrument is released as open hardware under CC BY-NC licence. The complete builder package — technical specification, custom parts register, machining instructions, bill of materials, calibration procedures, and builder's report template — is available at mumblejinx.github.io/opensource. The builder community is the peer review mechanism for this design. Each physical build that reports results through the Builder's Report Template converts an estimated confidence figure into a measured one.

References

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specification cited as precedent for VCE bearing design.]

Licence: This paper and all associated design documentation are released under the Creative Commons Attribution Non-Commercial licence (CC BY-NC). Anyone may build, study, modify, and share these designs. Commercial use requires explicit permission from the concept originator. Jake Galm, researched & documented 2026. Named for Alessandro Volta (1745–1827).